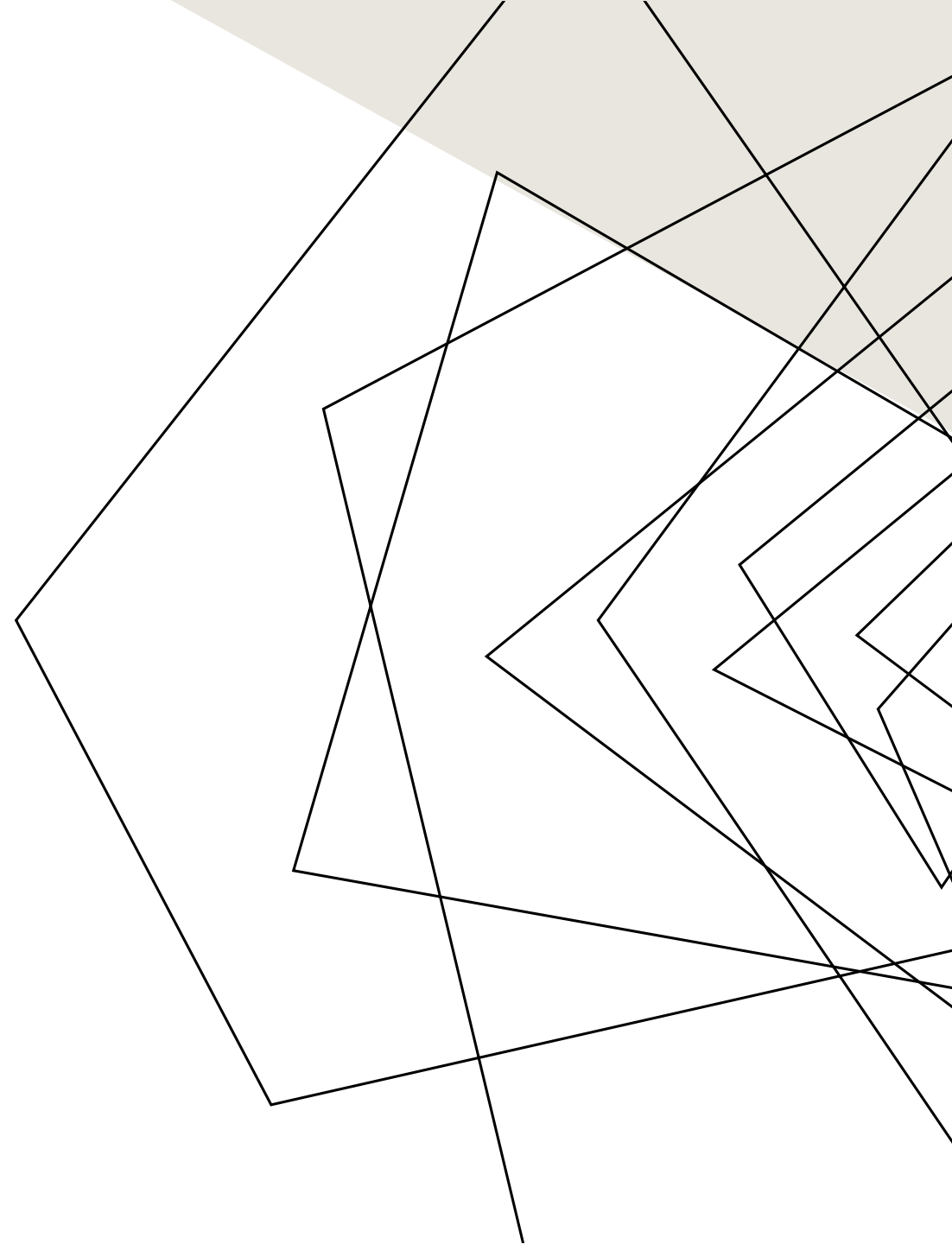


DATA SCIENCE CAPSTONE PROJECT

OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix



EXECUTIVE SUMMARY



METHODOLOGIES

- Data collection using we scraping from Wikipedia and using SpaceX api.
- Data wrangling including filling null spaces and merging data sources.
- Exploratory data analysis:
 - ✓ Visualizing the relationships between features of the data
 - ✓ SQL queries
 - ✓ Locating launch sites and marking point of interests on the map using folium
 - ✓ Creating an interactive dashboard that enables easy data analysis

ALL RESULTS

- Exploratory data analysis provides us with insights into the potential relationships between rocket launches and the features in the dataset.
- The correlation between the success rate and the payload mass was analyzed using an interactive dashboard.
- A machine learning model has been developed to predict whether SpaceX's first-stage landing will be successful or unsuccessful.
- Predictions made using various classification algorithms achieve accuracy rates of around 85%.



INTRODUCTION

PROBLEM

- A new commercial rocket launcher company
 - wants to bid against SpaceX
 - needs to advertise their rocket launches with a lower-cost
 - decided to analyze Falcon 9 launches to find a way
- Questions to be answered:
 - How do variables such as payload mass, launch site, number of flights, and orbits affect the success of the first stage landing?
 - The best model to predict whether the first stage lands successfully or not, which directly impacts on the cost
 - The best place to execute rocket launches



METHODOLOGY

Section I



PROBLEM

- Data collection methodology:
 - Requesting data using SpaceX's Rest API
 - web scrapping from Wikipedia's "List of Falcon 9 and Falcon Heavy Launches" page
- Perform data wrangling
 - Filtering the data
 - Dealing with missing data
 - Using One-Hot Encoding to prepare data for binary classification
- Perform exploratory data analysis (EDA) using visualization and SQL

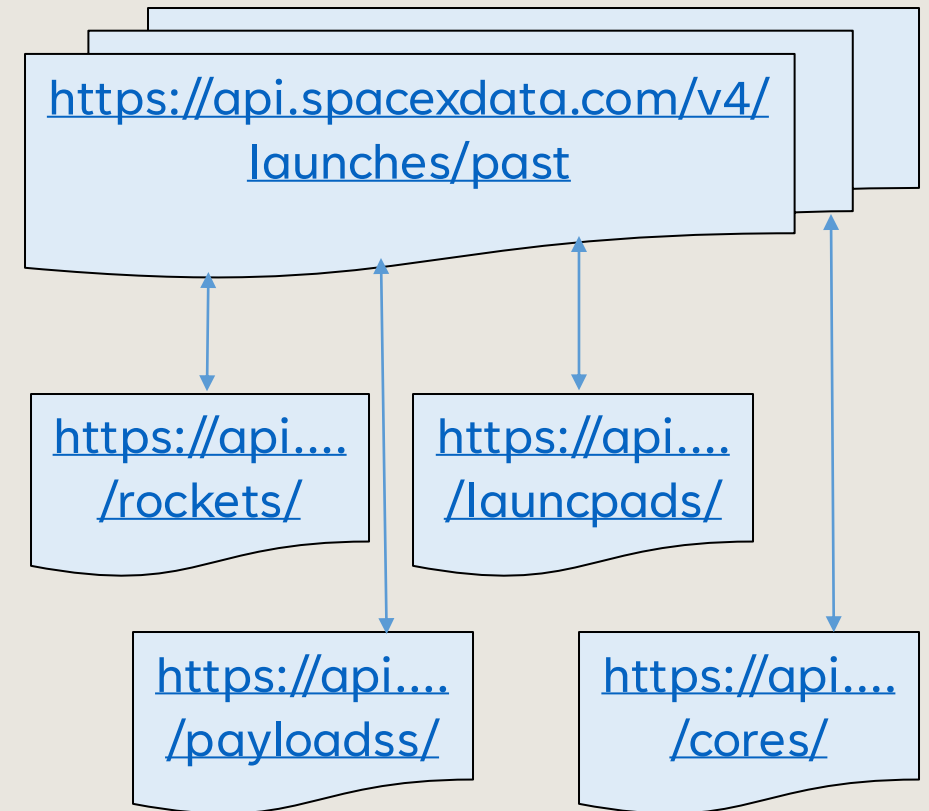
PROBLEM

- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - The data is standardized
 - Train-test split approach is used
 - The best hyperparameter for support vector machine (SVM), logistic regression, classification trees and k-nearest neighbors (KNN) is found
 - The best-performing method is obtained

DATA COLLECTION – SPACEX API

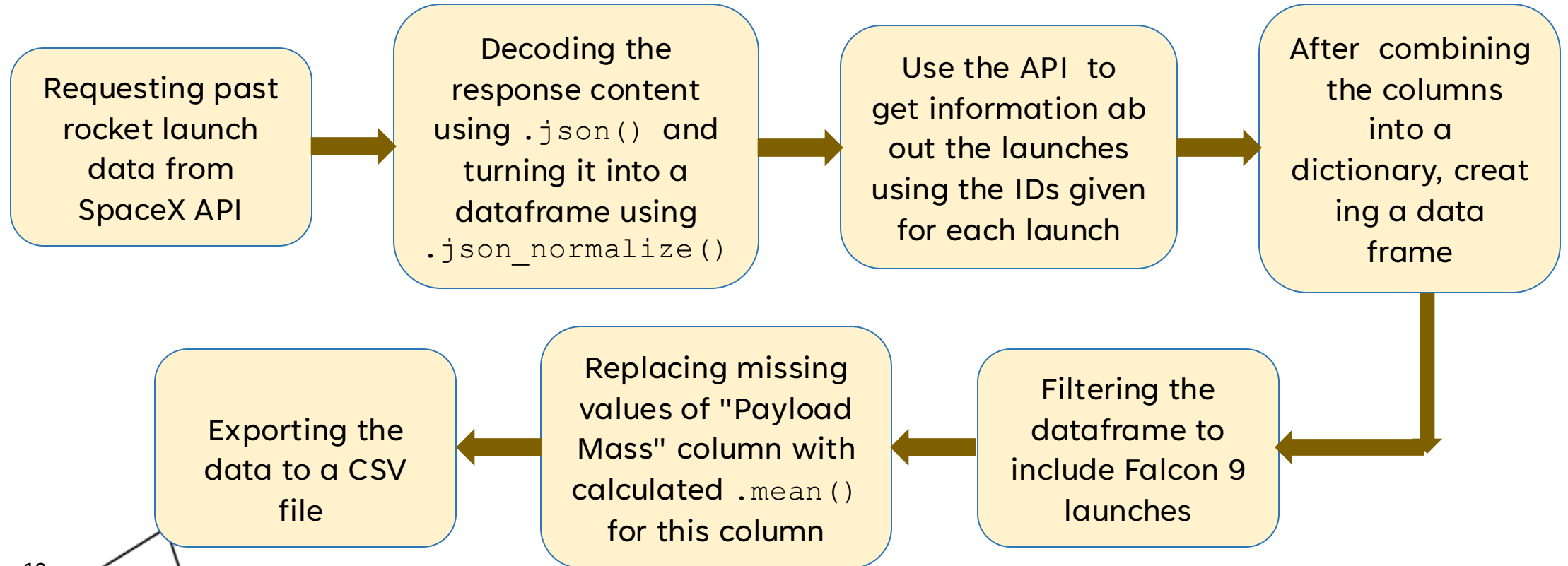
- Data collection process involved a combination of API requests from SpaceX REST API
- Web Scraping data from a table in SpaceX's Wikipedia entry for "List of Falcon 9 and Heavy Launches".

(https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922)



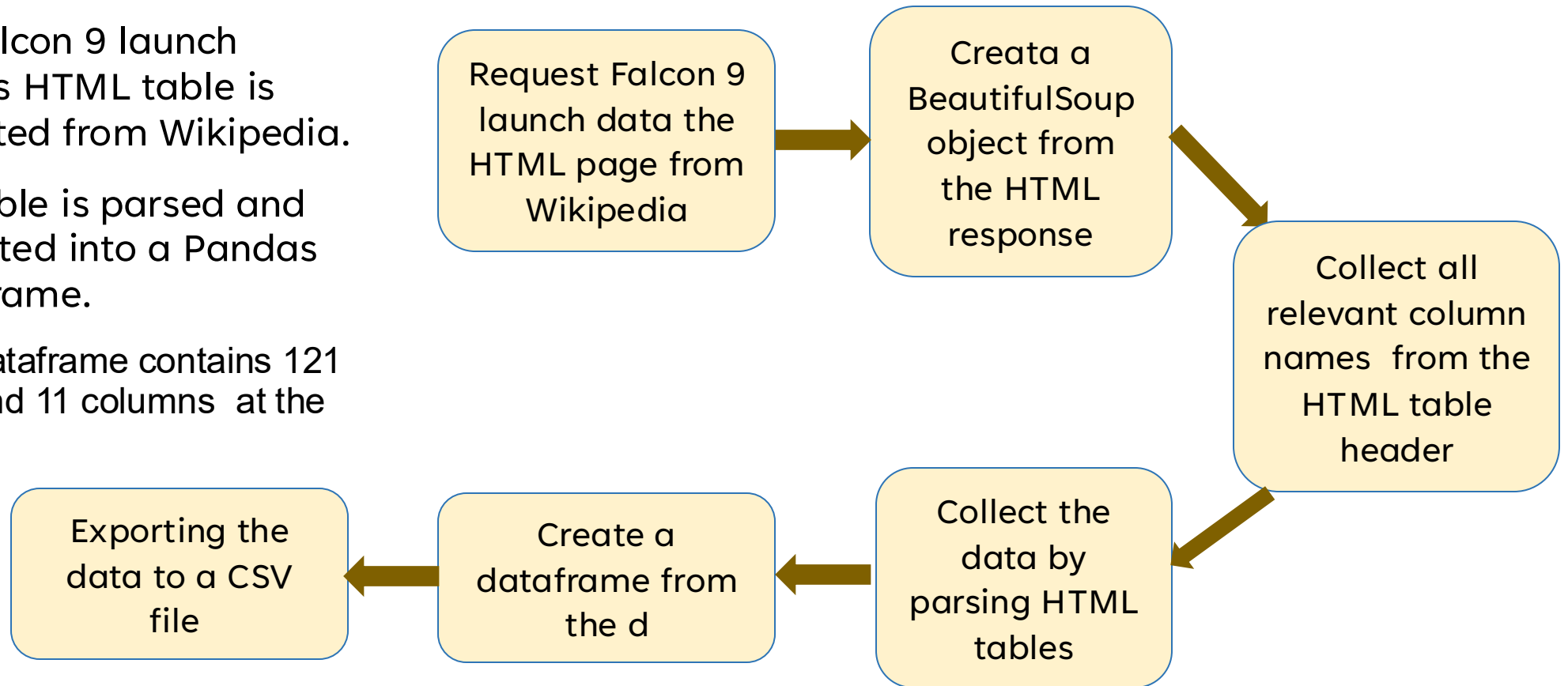
DATA COLLECTION – SPACEX API

- After the filtering process, the data frame contains 90 rows.



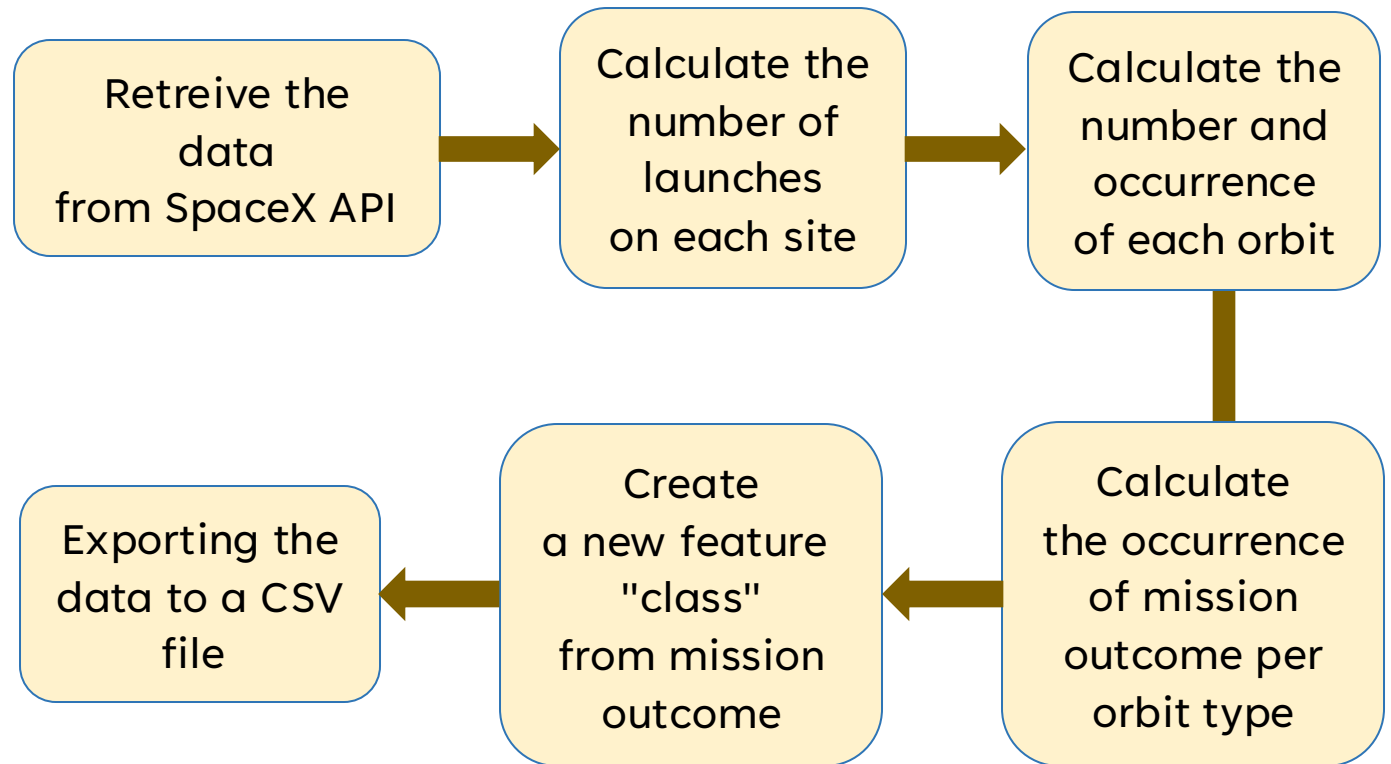
DATA COLLECTION – SCRAPING

- The Falcon 9 launch records HTML table is extracted from Wikipedia.
- The table is parsed and converted into a Pandas DataFrame.
- Final dataframe contains 121 rows and 11 columns at the end



DATA WRANGLING

- The dataset includes various scenarios in which the booster rocket failed to land successfully. Therefore, Exploratory Data Analysis was conducted to determine the training labels.
- These outcomes into Training Labels with “1” means the booster successfully landed, “0” means it was Unsuccessful.



EDA WITH DATA VISUALIZATION

- Various types of graphs and charts are created to analyze the relationships between data. These graphs are listed on the right.
- These graphs provide insights into how each key variable affects the success rate.
- Finally, categorical columns are converted into dummy variables for use in prediction models.
- **Scatter Plots:**
 - ❖ Flight Number vs. Payload Mass,
 - ❖ Flight Number vs. Launch Site,
 - ❖ Payload Mass vs. Launch Site,
 - ❖ Orbit Type vs. Success Rate,
 - ❖ Flight Number vs. Orbit Type,
 - ❖ Payload Mass vs Orbit Type and Success Rate Yearly Trend
- **Bar Graph:** Success rate of different orbit types
- **Line Graph:** Launch success rate yearly

EDA WITH SQL

- Performed SQL queries:
 - ❖ All launch sites for Falcon 9 are detected
 - ❖ Five records where launch sites begin with the string 'CCA' are displayed
 - ❖ The total payload mass carried by boosters launched by NASA (CRS) is printed
 - ❖ Average payload mass carried by booster version F9 v1.1 is calculated
 - ❖ The date of first successful landing for ground pad is gathered
 - ❖ The names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000 are listed
 - ❖ Mission outcomes are summarized
 - ❖ The names of the booster versions which have carried the maximum payload mass are listed
 - ❖ The failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015 are extracted
 - ❖ Landing outcomes and the number of them within specified dates are shown

BUILD AN INTERACTIVE MAP WITH FOLIUM

- **Markers & Circles**
 - **Clusters**
 - **Lines**
- MASA Johnson Space Center and launch sites are shown on the map with markers and circles
 - The number of launches is displayed by clustering them with marker clusters at each launch site
 - Launch sites and their distances from the mentioned locations are displayed on the map as colored lines

BUILD A DASHBOARD WITH PLOTLY DASH

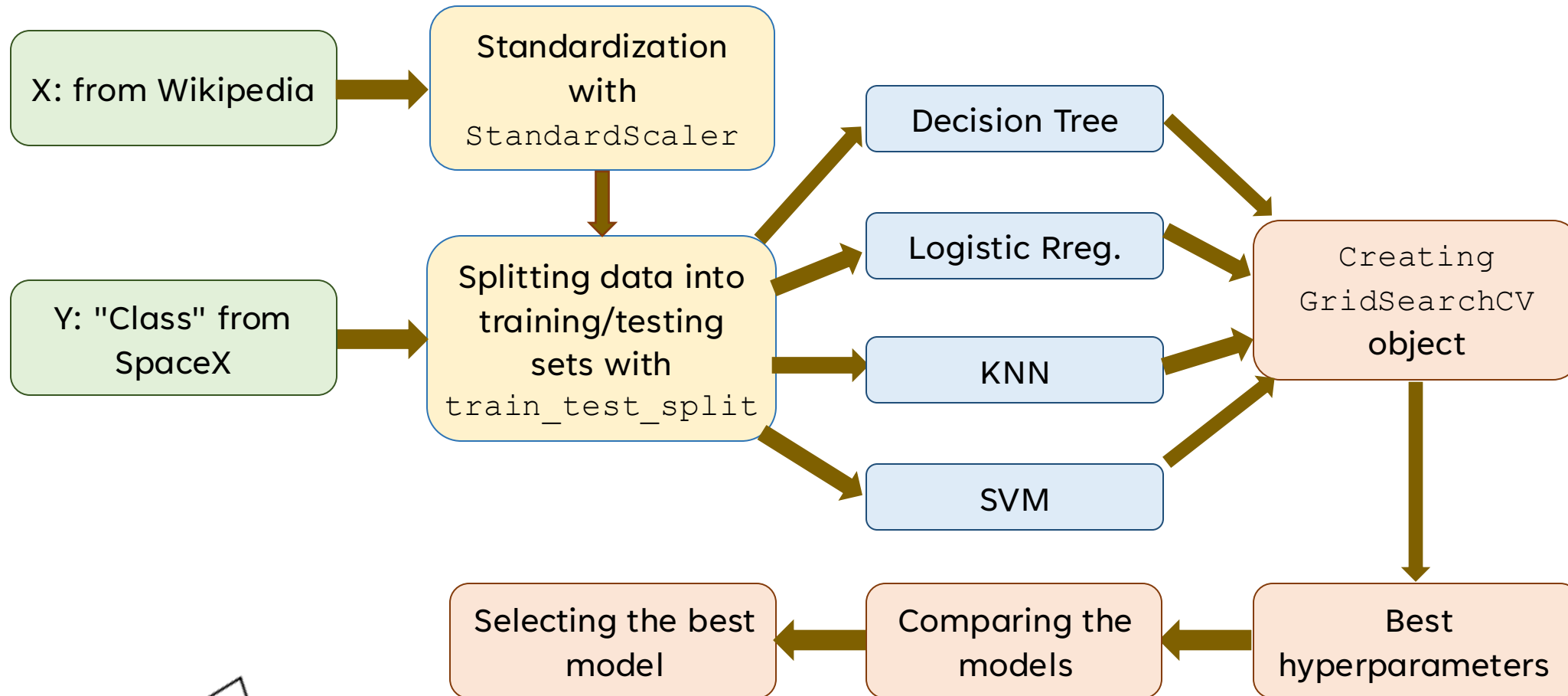
- Launch Sites Dropdown List:
 - ❖ A dropdown list to enable Launch Site selection
- Pie Chart showing Success Rate:
 - ❖ The success rate at each launch site is visualized. The launch sites are compared according to their successful first stage landing
- Slider of Payload Mass Range:
 - ❖ A slider to select Payload Mass (kg) range
- Scatter Chart:
 - ❖ The number of successful and failing landings within a specified payload mass is plotted with their booster versions

PREDICTIVE ANALYSIS – CLASSIFICATION

- X contains standardized version of the data fetched from Wikipedia
- Y contains the class information if first stage landed successfully from the data collected by calling SpaceX API
- 4 different classification models are built with SVM, KNN, logistic regression and classification trees.
- The best hyperparameters are found by grid searching
- Different models' accuracy on the test data are compared among each other to find the best model

[GitHub URL – Predictive Analysis](#)

PREDICTIVE ANALYSIS – CLASSIFICATION



RESULTS

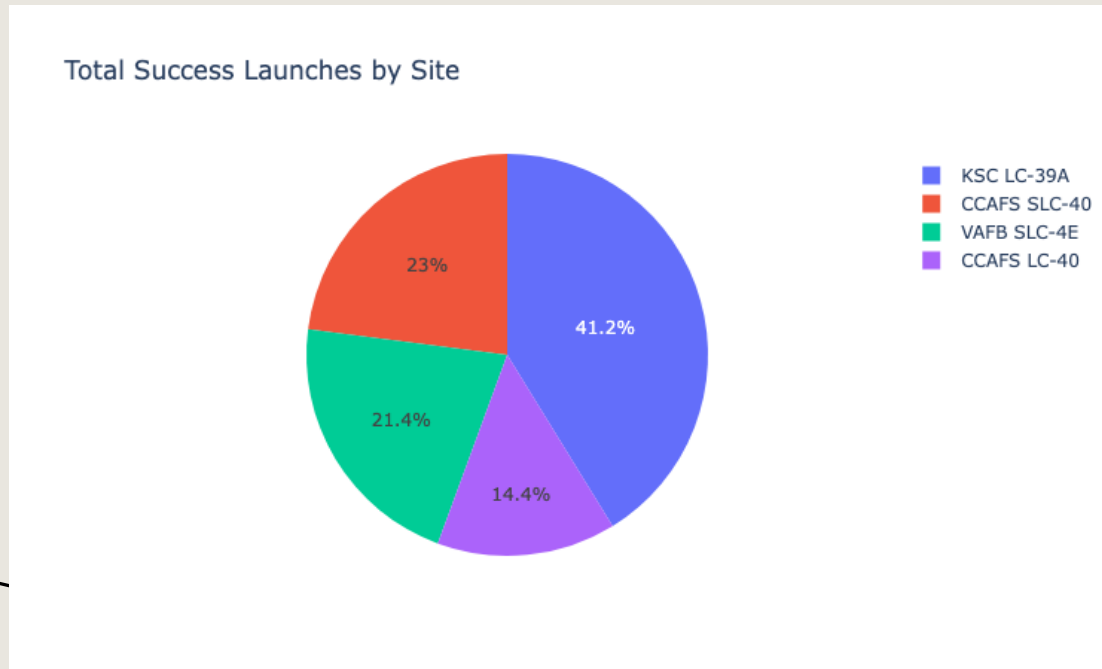
Exploratory Data Analysis Results

- SpaceX uses 4 different launch sites
- The average payload of F9 v1.1 is 2534 kg
- The success rate rises over the years
- Payload mass has different impacts on the outcome for different ranges
- Only 1 mission outcome is a failure
- Orbits do not affect the launch outcome

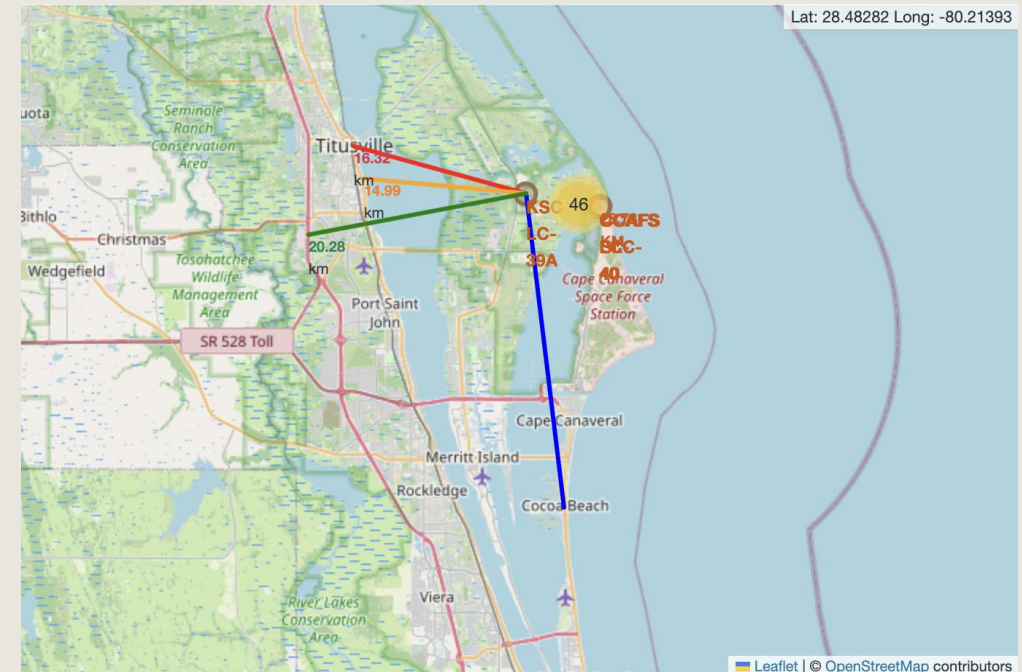
RESULTS

Interactive Analytics Demo in Screenshots

- An interactive dashboard is created



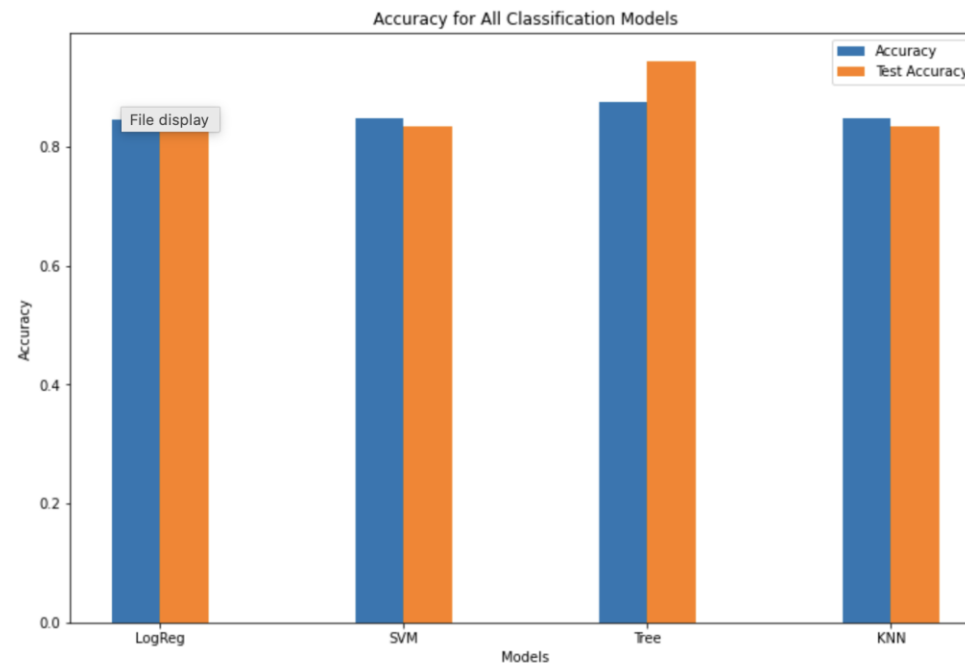
- Characteristics of launch sites are detected



RESULTS

Predictive Analysis Results

- Decision Tree is found as the best model performing 94% accuracy on the test data



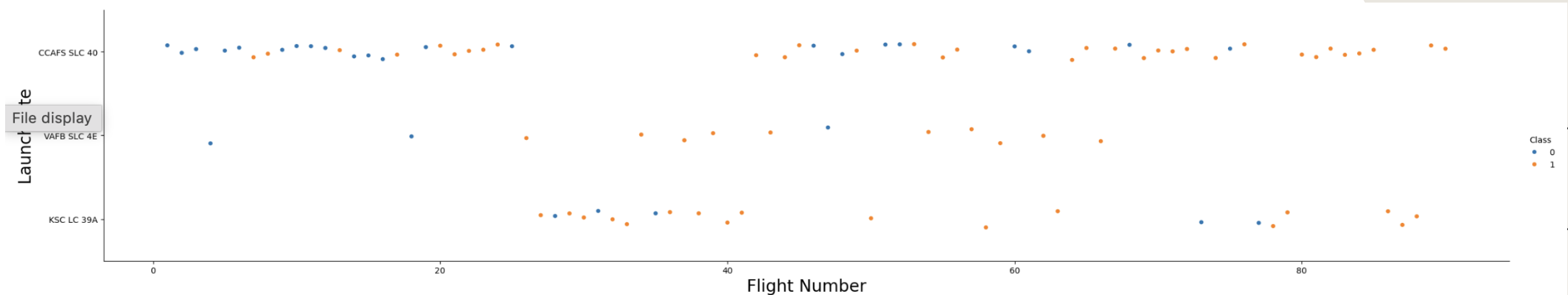
INSIGHTS DRAWN FROM EDA

Section II



EXPLORATORY DATA ANALYSIS RESULTS

Flight Number vs Launch Site

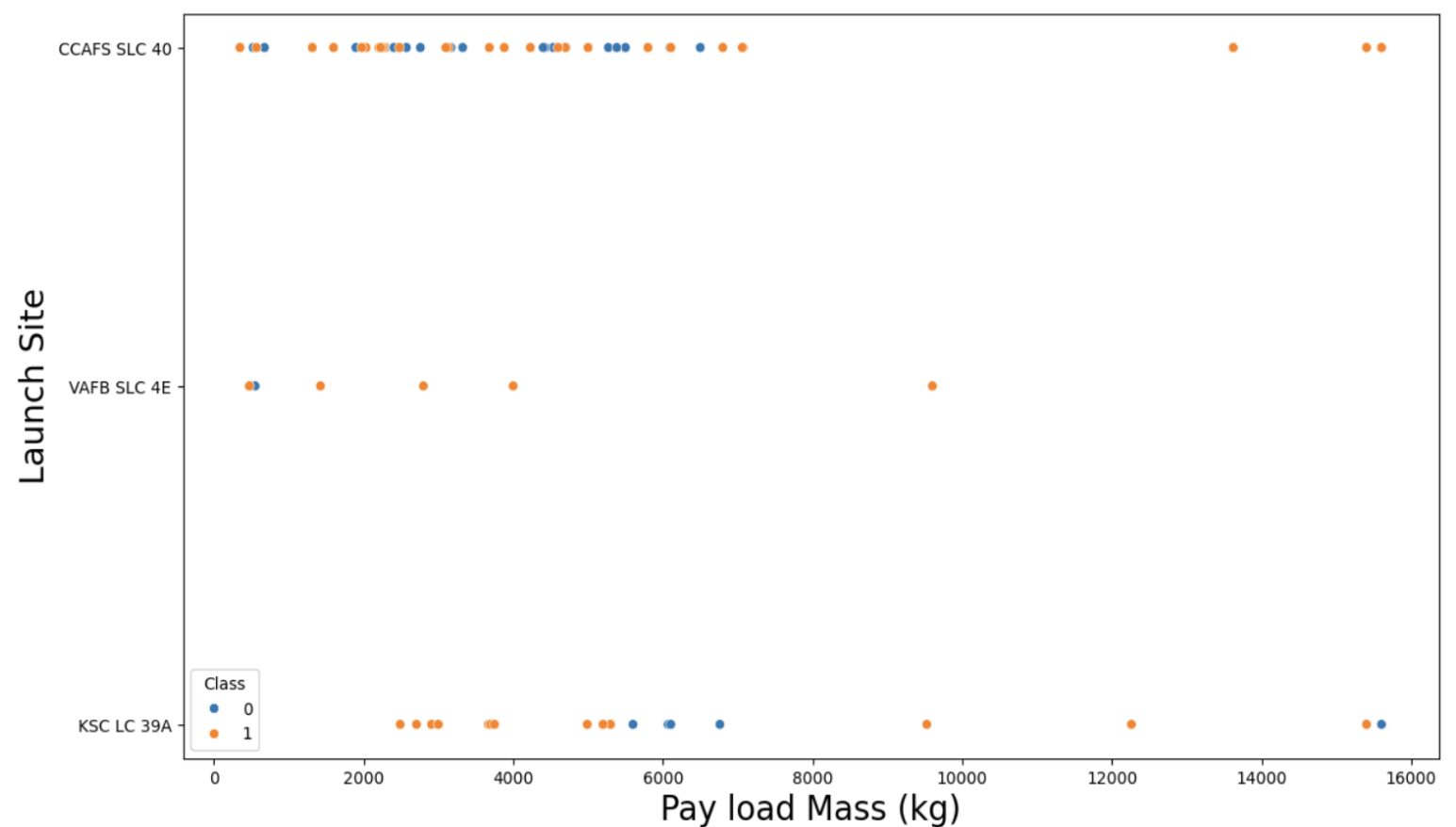


- While all the initial flights were unsuccessful, all the recent flights have been successful.
- The success rate rises over the years
- VAFB SLC 4E and KSC LC 39A have higher success rates.

EXPLORATORY DATA ANALYSIS RESULTS

Payload vs Launch Site

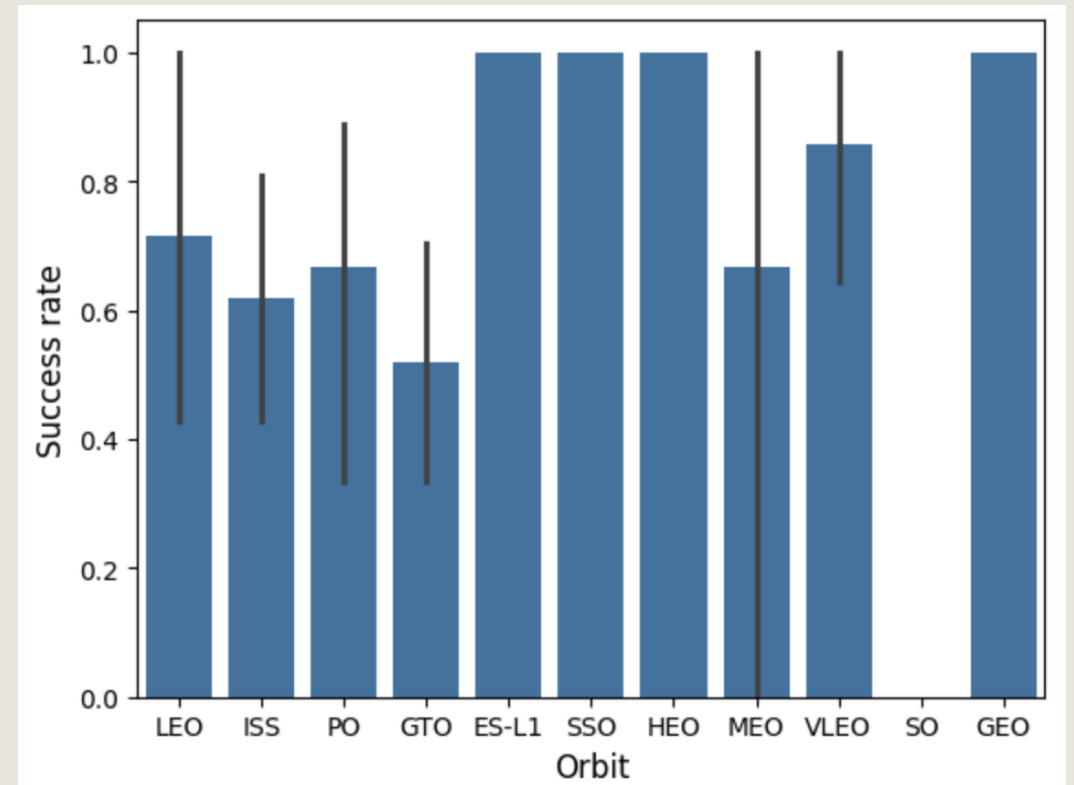
- Launches exceeding 10,000 kg are not conducted from VAFB SLC-4E.
- The payload mass of rockets typically ranges from 1,000 to 8,000.
- Launches with a payload mass between 8,000 and 15,000 kg are quite rare.
- Launches exceeding 7,000 kg carry less risk.



EXPLORATORY DATA ANALYSIS RESULTS

Success rate vs. Orbit type

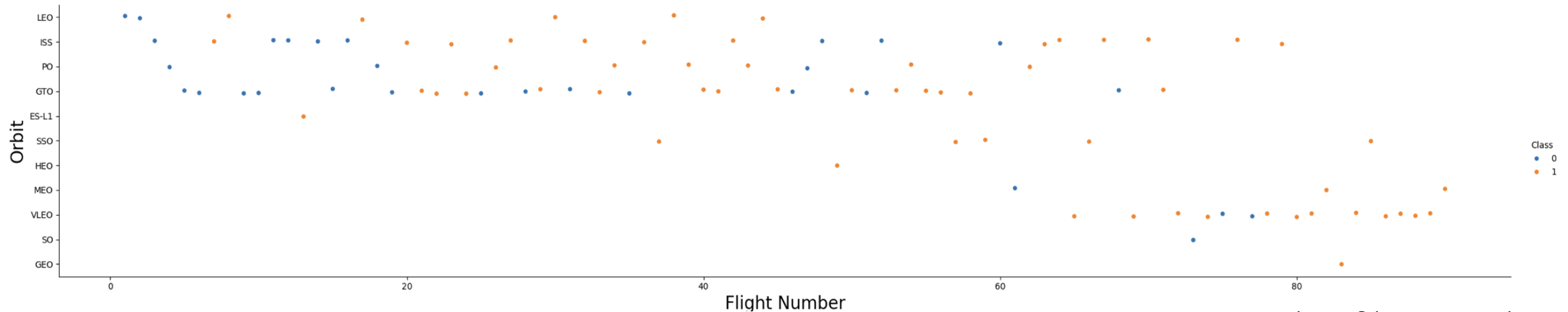
- Orbits with 100% success rate: ES-L1, GEO, HEO, SSO
- Orbits with 0% success rate: SO
- Orbits with success rate between 50% and 85%: GTO, ISS, LEO, MEO, PO



EXPLORATORY DATA ANALYSIS RESULTS

Flight Number vs Orbit Type

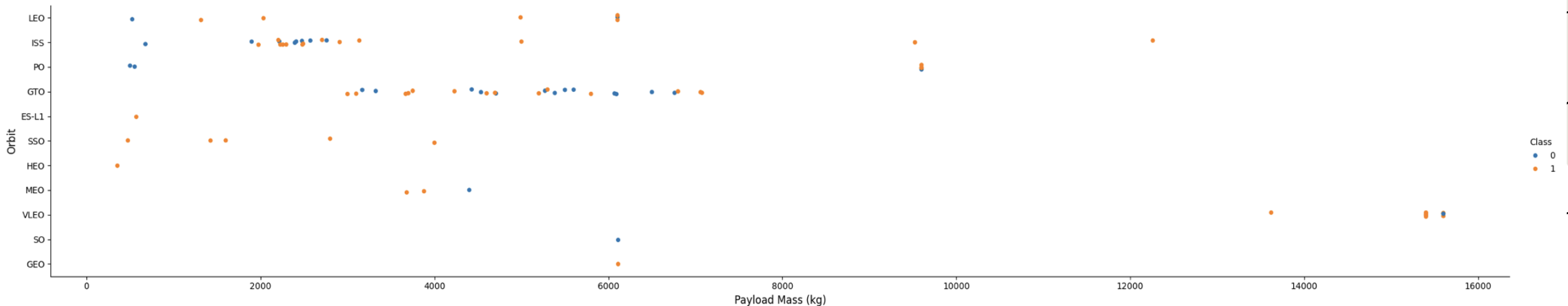
- While success in Low Earth Orbit (LEO) appears to be correlated with the number of flights, no such relationship is found for Geostationary Transfer Orbit (GTO)



EXPLORATORY DATA ANALYSIS RESULTS

Payload vs Orbit Type

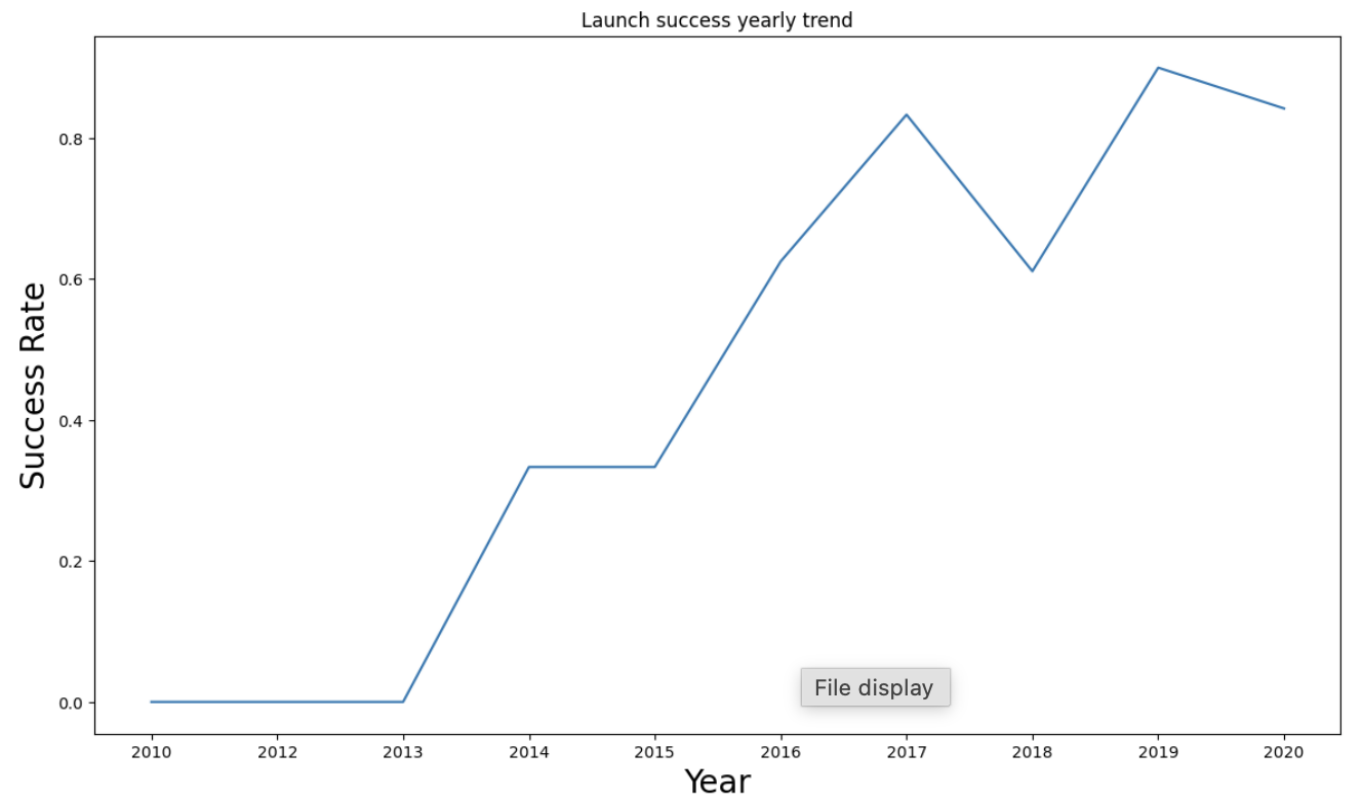
- The rate of successful or positive landings with heavy loads is higher for Polar, LEO, and the ISS.
- However, in the case of GTO, because both outcomes are possible, it is difficult to distinguish successful from unsuccessful landings.



EXPLORATORY DATA ANALYSIS RESULTS

Launch Success Yearly Trend

- The success rate has continued to increase since 2013 until 2020.



EXPLORATORY DATA ANALYSIS RESULTS

All Launch Sites

- There are four launch sites used for space missions.

```
%sql select distinct Launch_Site from SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

EXPLORATORY DATA ANALYSIS RESULTS

Launch Sites Names Begin with "CCA"

- 5 records where launch sites begin with the string 'CCA' are displayed

```
%sql select * from SPACEXTABLE where Launch_Site like "CCA%" limit 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

EXPLORATORY DATA ANALYSIS RESULTS

Total Payload Mass

- The total payload mass carried by boosters launched by NASA (CRS) is calculated as

```
%sql select sum(PAYLOAD_MASS__KG_) as Total_payload_mass from SPACEXTABLE where customer = "NASA (CRS)";
```

```
* sqlite:///my_data1.db  
Done.
```

Total_payload_mass

File display 45596

EXPLORATORY DATA ANALYSIS RESULTS

Average Payload Mass by F9 v1.1

- The average payload mass carried by boosters F9 v1.1 is calculated as

```
%sql select avg(PAYLOAD_MASS__KG_) as Average_payload_mass from SPACEXTABLE where Booster_Version like "%F9 v1.1%";
```

```
* sqlite:///my_data1.db  
Done.
```

<u>Average_payload_mass</u>

2534.6666666666665

EXPLORATORY DATA ANALYSIS RESULTS

First Successful Ground Landing Date

- The date when the first succesful landing outcome in ground pad is found as

```
%sql select min(Date) as first_successful_landing from SPACEXTABLE where Landing_Outcome = "Success (ground pad)";
```

```
* sqlite:///my_data1.db  
Done.
```

<u>first_successful_landing</u>

2015-12-22

EXPLORATORY DATA ANALYSIS RESULTS

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of boosters that have successfully landed on a drone ship and had a payload mass greater than 4000 but less than 6000

```
%sql select Booster_Version from SPACEXTABLE where Landing_Outcome = "Success (drone ship)" and PAYLOAD_MASS_KG_
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

EXPLORATORY DATA ANALYSIS RESULTS

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes are found as

```
%sql select Mission_Outcome, count(*) as Total_number from SPACEXTABLE group by Mission_Outcome;
```

```
* sqlite:///my_data1.db  
Done.
```

Mission_Outcome	Total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EXPLORATORY DATA ANALYSIS RESULTS

Boosters Carried Maximum Payload

- The names of the launch vehicles capable of carrying the highest payload mass are listed below.

```
%sql select Booster_Version from SPACEXTABLE where PAYLOAD_MASS_KG_
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

EXPLORATORY DATA ANALYSIS RESULTS

2015 Launch Records

- The unsuccessful landings that occurred on the uncrewed landing outcomes in 2015, along with the relevant booster versions and launch site names, are as follows:

```
%sql select substr(Date, 6,2) as Month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE where
```

```
* sqlite:///my_data1.db
```

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

EXPLORATORY DATA ANALYSIS RESULTS

Rank Success Count between 2010-06-04 and 2017-03-20

```
%sql select Landing_Outcome, count(*) as Count_outcomes from SPACEXTABLE where Date between
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	Count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- The counts of landing results between June 4, 2010, and March 20, 2017, are listed in descending order.

A hand is shown on the left side of the image, pointing towards the right. The background is a dark blue gradient with white circuit-like patterns. A small, glowing blue robot head icon is visible in the lower-left quadrant. The right side of the image is black with white geometric lines forming a grid-like pattern.

LAUNCH SITES PROXIMITIES ANALYSIS

Section III

LOCATIONS OF ALL LAUNCH SITES ON THE MAP

- The map shows VAFB SLC-4E, CCASF (S)LC-40, and KSC LC-39A locations.
- While KSC LC-39A and CCASF (S)LC-40 are located very close to each other, VAFB SLC-4E is situated on the western side of the country.



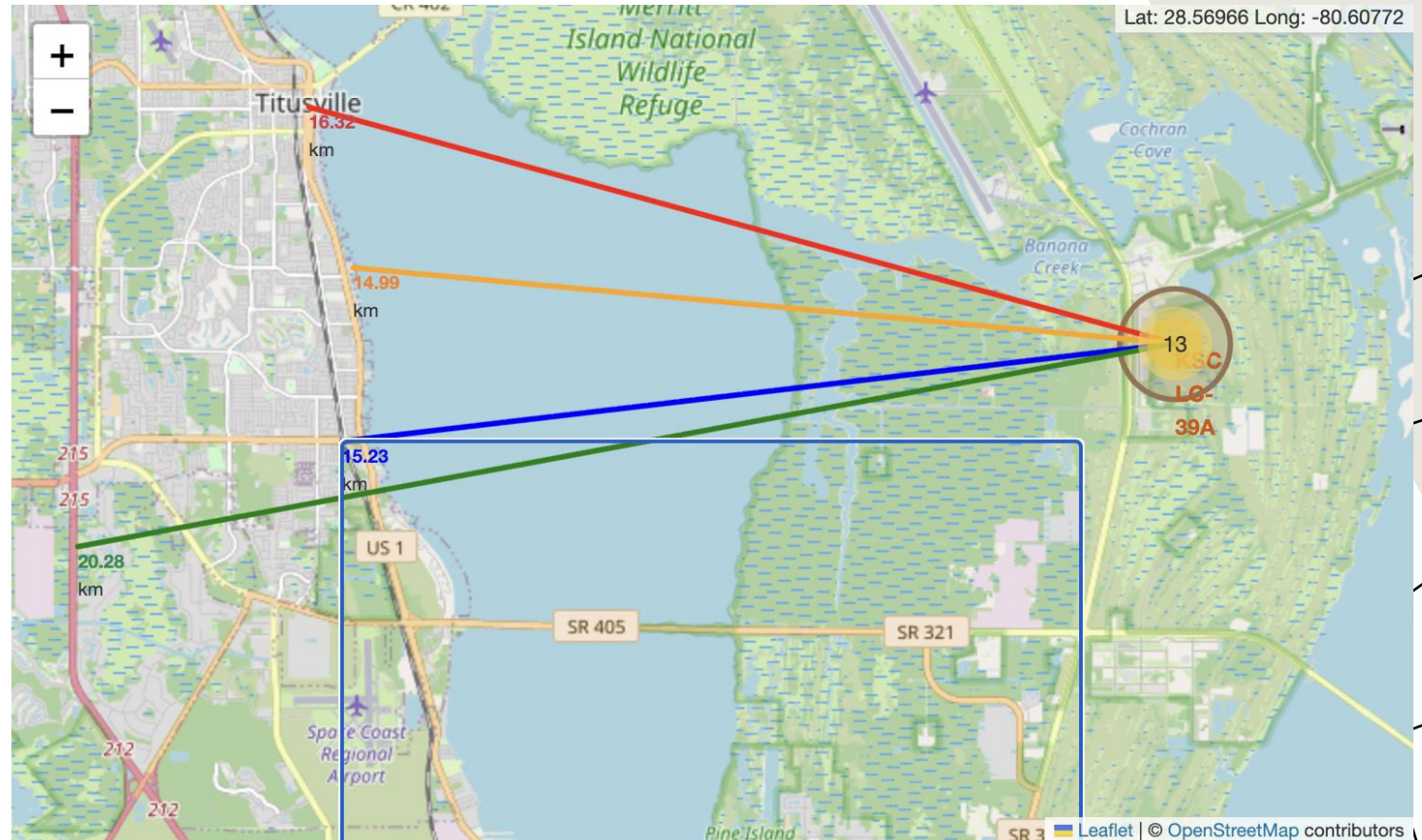
LOCATIONS OF ALL LAUNCH SITES ON THE MAP

- The map shows VAFB SLC-4E, CCASF (S)LC-40, and KSC LC-39A locations.
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LOCATIONS OF ALL LAUNCH SITES ON THE MAP

- The map shows VAFB SLC-4E, CCASF (S)LC-40, and KSC LC-39A locations.
- While KSC LC-39A and CCASF (S)LC-40 are located very close to each other, VAFB SLC-4E is situated on the western side of the country.



BUILD A DASHBOARD WITH PLOTLY DASH

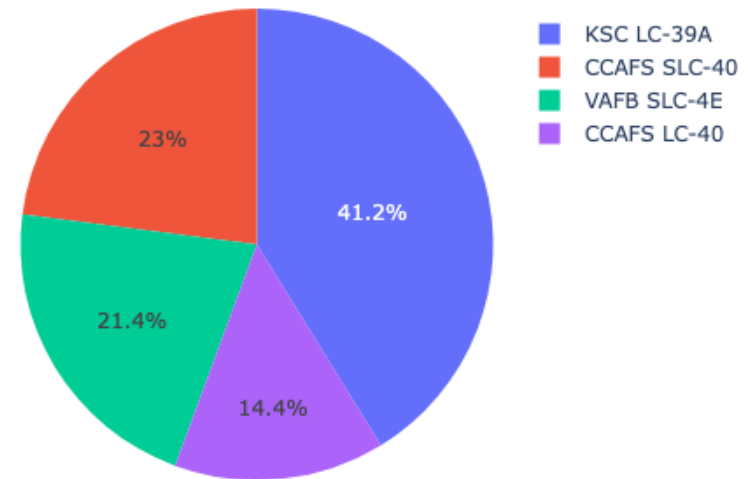
Section IV



SUCCESS RATIO – ALL SITES

- KSC LC-39A has the highest success ratio among others, followed by CCAFS SLC-40

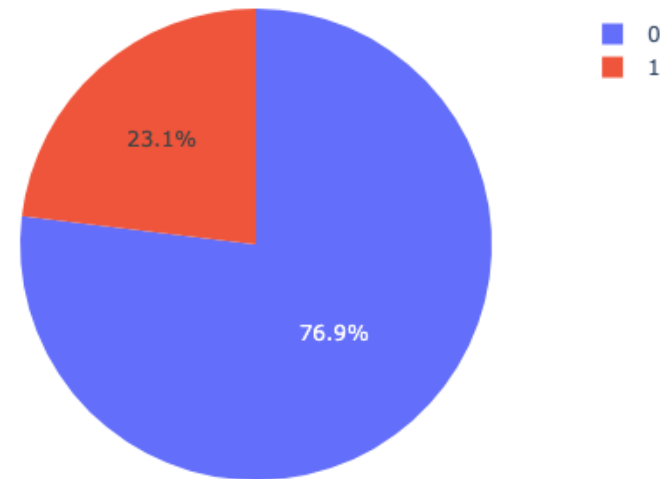
Total Success Launches by Site



SUCCESS RATIO – BEST SITE

- The highest success ratio, which is 76.9% with 10 successful and only 3 failed landings, has been achieved in KSC LC-39A launch site.

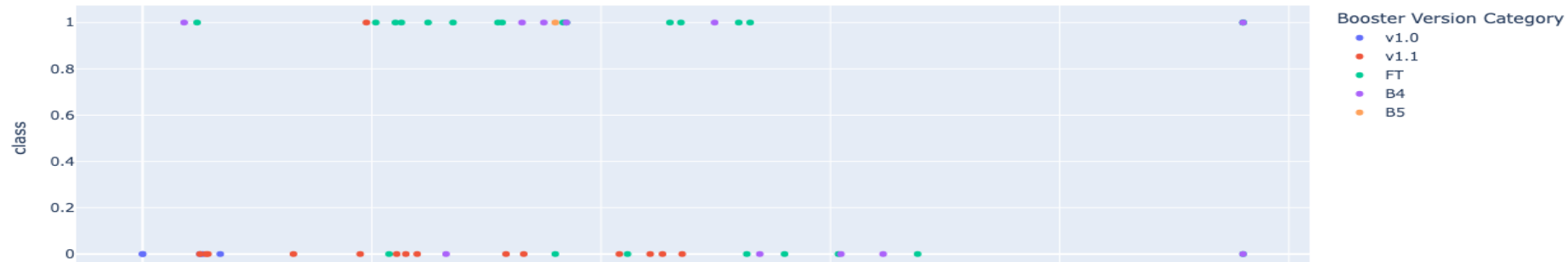
Total Success Launches for Site KSC LC-39A



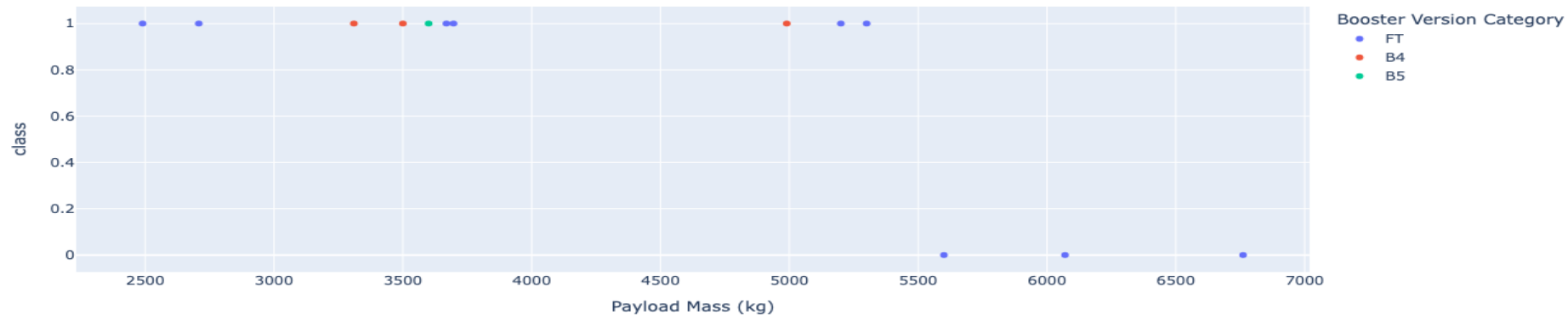
CORRELATION

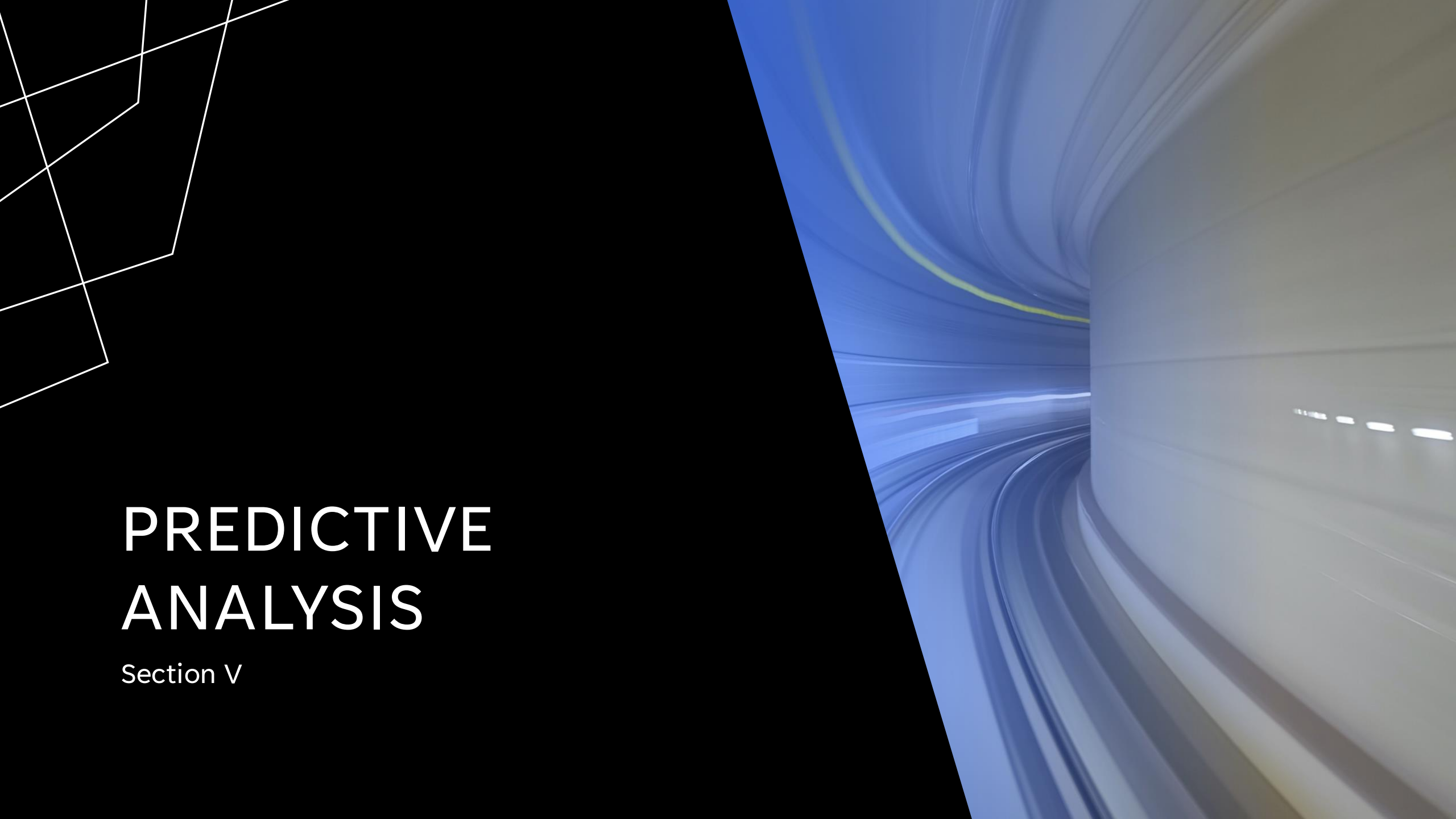
- The graphs show that payloads between 2,000 and 5,500 kg have the highest success rate.

Correlation Between Payload Mass and Success for All Sites



Correlation Between Payload Mass and Success for Site KSC LC-39A





PREDICTIVE ANALYSIS

Section V

CLASSIFICATION ACCURACY

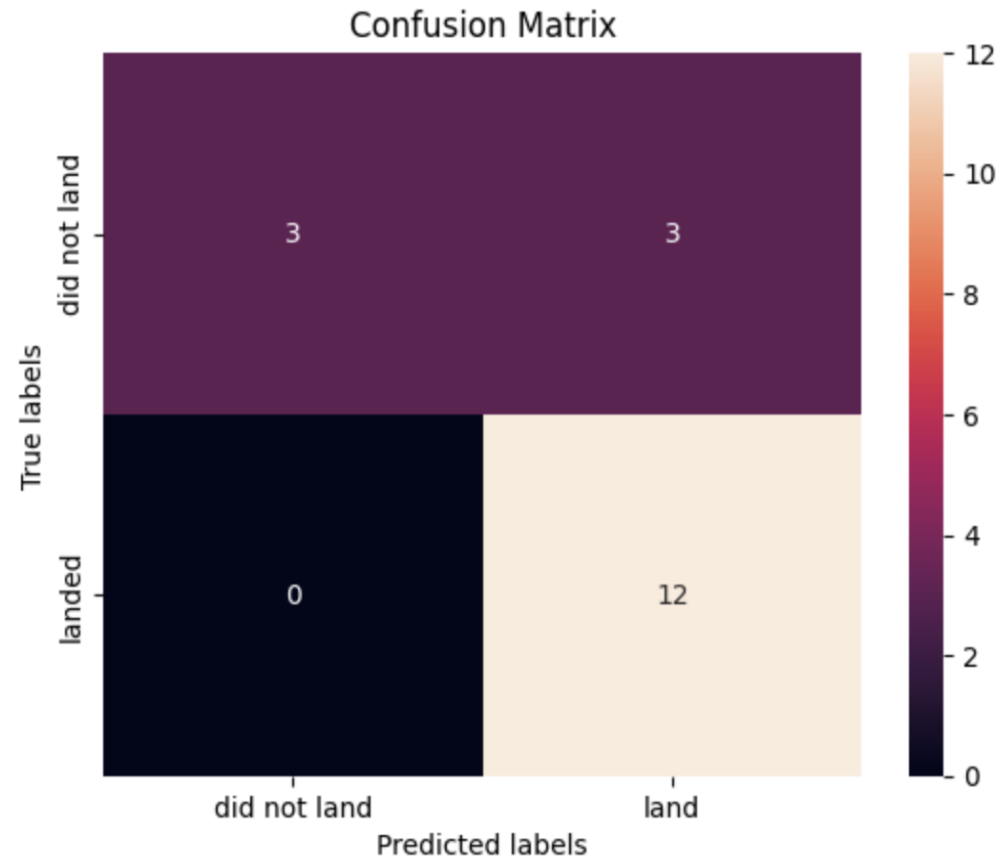
- Based on the test set results, we cannot confirm which method performed the best.
- The scores for the entire dataset confirm that the best model is the Decision Tree Model.

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.800000	0.800000	0.800000	0.800000
F1_Score	0.888889	0.888889	0.888889	0.888889
Accuracy	0.833333	0.833333	0.833333	0.833333

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.833333	0.845070	0.842857	0.819444
F1_Score	0.909091	0.916031	0.914729	0.900763
Accuracy	0.866667	0.877778	0.877778	0.855556

CONFUSION MATRIX

- When examining the confusion matrix, we see that the decision tree can distinguish among the different classes. The main issue appears to be false positives.



CONCLUSIONS

- Most launch sites are located close to the Equator, and all of them are situated very close to the coast
- The success rate of launches has increased over the years.
- KSC LC-39A has the highest launch success rate among all launch sites.
- Orbits ES-L1, GEO, HEO and SSO have 100% success rate.
- Launches with low payload mass yield better results compared to launches with higher payload mass.
- Decision Tree Model is the best classification algorithm for this dataset.

APPENDIX

- Folium does not display maps in .ipynb files on GitHub. Therefore, the maps can only be accessed via this report presentation file.